

Aditya Rana

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[LinkedIn](#) • [Github](#) • [Leetcode](#) • [Personal Portfolio](#)



PROFESSIONAL SUMMARY

Aspiring AI/ML Engineer with hands-on experience in machine learning and modern AI frameworks like LangGraph, LangChain, and n8n. Strong foundation in data structures and algorithms (200+ LeetCode problems) and experience with RAG, embeddings, and LLM-based applications. Aspiring to build scalable and impactful AI-driven systems.

EDUCATION

Bachelor of Technology (Artificial intelligence)	2024 - 2028
Newton School of Technology,Rishihood University	Grade: 7.2/10.0
Intermediate (Class XII)	2023 - 2024
Modern Sandeepni School	Grade: 82.6%
Matriculation (Class X)	2021 - 2022
Modern Sandeepni School	Grade: 91.0%

INTERNSHIPS

Frontend Developer Intern	June 2025 - July 2025
TechQRT	remote
- Built a React-based UI for a finance broker app with responsive layouts and smooth user experience. - Integrated REST APIs to display real-time financial data. - Collaborated via GitHub for version control and team workflows. - Gained hands-on experience in React.js, API integration, and scalable frontend design.	

PROJECTS

exam-ai-assistant (Github) (Demo)	April 2026
Developed ExamMaster AI, a high-performance RAG-based platform using FastAPI and React to convert study materials into interactive knowledge systems. Built a semantic search pipeline with LangChain, FAISS, and HuggingFace embeddings for context-aware answers, revision notes, and mock exams. Designed an interactive UI with advanced animations and features like flashcards and syllabus tracking. Optimized performance with model pre-loading and structured outputs via OpenRouter for seamless integration.	
Cineraama (Github) (Demo)	April 2026
Developed a high-performance Movie Recommendation Engine featuring a decoupled microservice architecture with a FastAPI Python backend and a Streamlit frontend, engineered with a premium dark-themed "Midnight Cinema" UI using raw CSS injection for an immersive, sophisticated user experience. Architected an advanced content-based filtering system using Scikit-Learn to compute Term Frequency-Inverse Document Frequency (TF-IDF) cosine similarities across extensive movie attributes, instantly delivering highly accurate and personalized "Because you liked" film suggestions. Engineered a real-time metadata enrichment pipeline integrating the live TMDB API to dynamically fetch high-resolution posters, audience metrics, and synopses, while optimizing backend cold starts by efficiently deserializing pre-computed, multi-gigabyte machine learning models in production.	
EV DEMAND FORECASTING SYSTEM (Github) (Demo)	March 2026
- Built an EV Charging Demand Forecasting System with a dual architecture (Streamlit prototype + FastAPI/React production). - Developed an ensemble ML pipeline using Gradient Boosting models for accurate demand prediction. - Implemented LangGraph-based agent planning with RAG using FAISS and OpenRouter for infrastructure recommendations.	

- Applied advanced **feature engineering and production optimizations** (time-series features, async loading, background processing).

SKILLS

Computer Languages: Machine Learning, TypeScript, HTML, CSS, JavaScript, Python, SQL

Data Tools: PostgreSQL, Power BI, TensorFlow, NumPy

Software Packages: Pandas, Matplotlib, Figma, Tailwind, Tailwind CSS, Next JS, Prisma ORM, React, Node.js, Excel, Express JS, MySQL, MongoDB

Additional Courses: Data Structure

Soft Skills: Team Building, Time management, Teamwork, Critical Thinking, Responsibility, People skills, Communication Skills

Others: Git and Github, Firebase, Email Marketing, Microsoft Office, Spreadsheet, Canva, Analytics, GenAI, GitHub Actions, Hugging Face, Framer Motion, OAuth 2.0, Tableau, Generative AI

EXTRA-CURRICULAR ACTIVITIES

- **District-Level Badminton Player**
- **State-Level Weightlifter**
- **District-Level Group Song Winner**
- **Swimming and Gaming Enthusiast**
- **Active Member of the Dev Club, Newton School of Technology**